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Predicting Income Levels Using Machine Learning Models

Introduction

This project focuses on predicting whether an individual earns more than $50,000 per year using demographic and employment-related data from the Adult dataset. The dataset includes information such as age, education level, occupation, hours worked per week, and other personal attributes that may influence income level.

The goal of this analysis is to build and evaluate machine learning models that can accurately classify individuals into two categories: those earning more than $50,000 annually and those earning less. This is a binary classification problem, where the target variable, income, is converted into numerical values for modeling purposes.

To accomplish this, three different classification models were implemented: Logistic Regression, Decision Tree, and Random Forest. These models were chosen to compare how different approaches, including linear, rule-based, and ensemble-based methods, perform on the same dataset.

The performance of each model is evaluated using accuracy and F1 score, which allows for a clearer comparison of their effectiveness. In addition to evaluating performance, this report also examines which features were most influential in predicting income and discusses the key patterns observed in the data.

Methodology

To predict whether an individual earns more than $50,000 per year, the Adult dataset was first preprocessed and prepared for the machine learning models. The dataset contains both numerical and categorical features, so several modifications were made to ensure the data could be used properly.

The target variable, income, was converted into binary format, where individuals earning less than or equal to $50,000 were labeled as 0 and those earning more than $50,000 were labeled as 1. This allowed the problem to be treated as a binary classification task.

Before training the models, the dataset was split into training and testing sets using an 80/20 split. The training set was used to fit the models, while the testing set was used to evaluate their performance on unseen data.

Since the dataset includes categorical variables such as workclass, education, and occupation, these features were transformed using one-hot encoding. This process converts categorical data into numerical columns so that machine learning algorithms can interpret them. In addition, missing values in the dataset were handled using imputation, where numerical values were filled with the median and categorical values were filled with the most frequent category.

For numerical features, standardization was applied in the Logistic Regression model to ensure that all features were on a similar scale. This step was important so that larger values did not dominate the model. However, scaling was not applied to the Decision Tree and Random Forest models because these models are not affected by differences in feature scale.

Three different classification models were used: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was used as a baseline model because of its simplicity and interpretability. The Decision Tree model was used to capture nonlinear relationships by partitioning the data into similar groups using if-then-else rules. Finally, the Random Forest model, which is an ensemble of multiple decision trees, was used to improve performance and reduce overfitting.

Each model was evaluated using accuracy and F1 score. Accuracy measures the overall proportion of correct predictions, while F1 score provides a balance between precision and recall. F1 score was preferred over using precision or recall individually because it accounts for the tradeoff between the two and gives a more complete picture of how the model is performing.

Findings

The performance of the three models, Logistic Regression, Decision Tree, and Random Forest, was evaluated using accuracy and F1 score. Overall, all three models achieved similar levels of performance, with accuracy values ranging from approximately 0.85 to 0.86. As shown in Figure 1, the differences between the models are relatively small across both metrics.

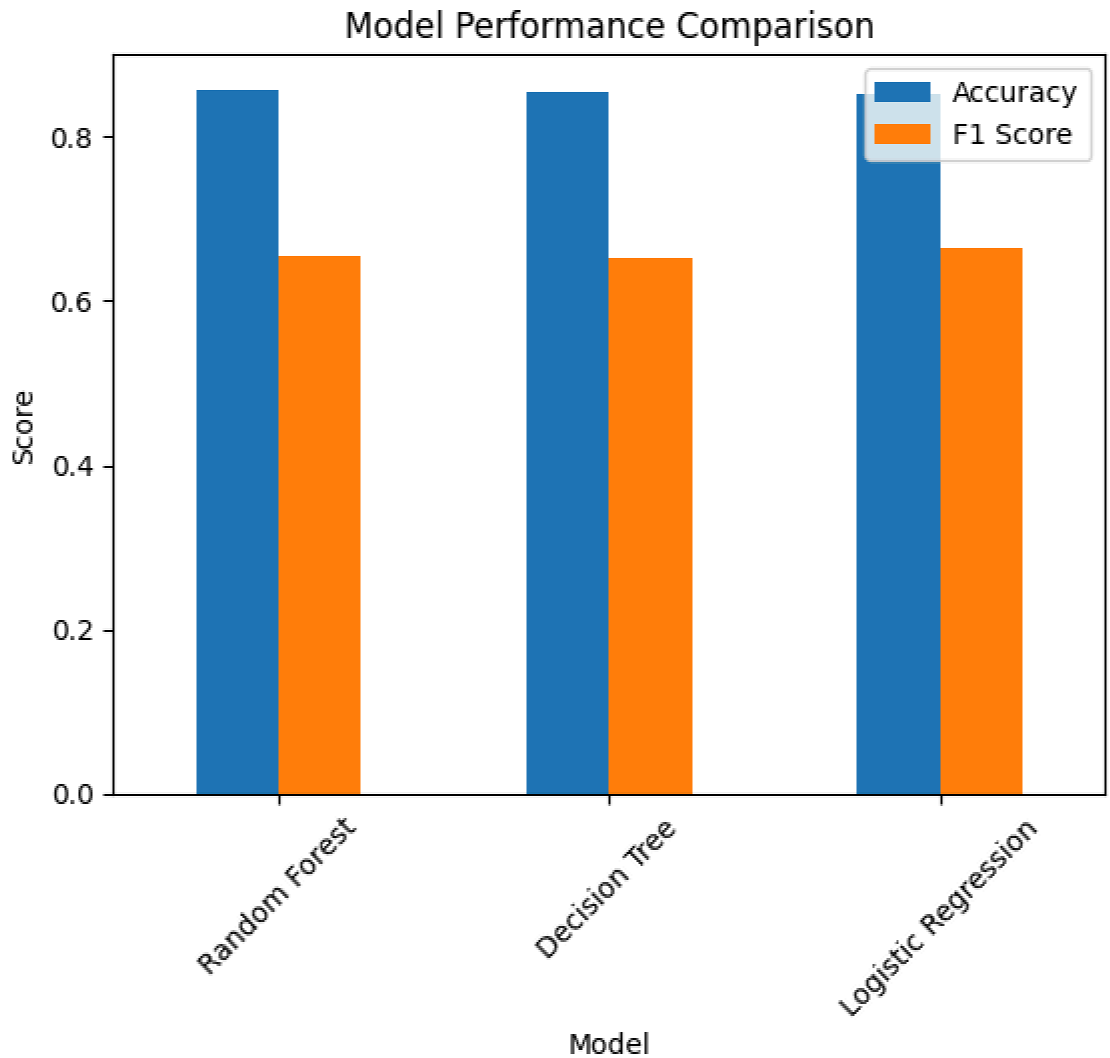

Figure 1: Model Performance Comparison (Accuracy and F1 Score)

The Random Forest model achieved the highest accuracy at 0.8575, followed closely by the Decision Tree at 0.8552 and Logistic Regression at 0.8531. However, when comparing F1 scores, Logistic Regression performed slightly better at 0.6636, compared to 0.6550 for Random Forest and 0.6532 for Decision Tree. This suggests that although Random Forest had the highest overall accuracy, Logistic Regression provided a more balanced performance between precision and recall.

Across all models, a consistent pattern emerges in the classification reports. Each model performs significantly better at predicting the lower-income class (≤ 50K) than the higher-income class (> 50K). For example, recall for the higher-income class ranges from 0.56 to

0.60 across the models, indicating that a substantial number of high-income individuals are misclassified as low-income.

This trend is also reflected in the confusion matrices. All three models correctly classify the majority of low-income individuals but struggle more with identifying high-income individuals. For instance, the Random Forest model misclassified 687 high-income individuals, while the Decision Tree and Logistic Regression misclassified 680 and 624, respectively. This suggests that all models are influenced by the class imbalance in the dataset and tend to favor the majority class.

Key Features

Feature importance from both the Decision Tree and Random Forest models provides insight into which variables were most influential in predicting income. While the exact rankings differ slightly between the models, several key features consistently appear as strong predictors, including marital status, education level, capital gain, age, and hours worked per week.

In the Decision Tree model, marital status, specifically being married, was the most important feature by a large margin, followed by education level and capital gain. This suggests that the model relies heavily on a few clear rule-based splits, where certain characteristics strongly separate the income groups. In contrast, the Random Forest model distributes importance more evenly across features, with capital gain, marital status, and education level still ranking highly, but with additional emphasis on relationship status and age.

Capital gain stands out as a key financial indicator in both models, suggesting that individuals with investment-related income are more likely to fall into the higher-income category. Similarly, education level remains a strong predictor, reinforcing the relationship between higher education and earning potential. Marital and relationship-related features, such as being married or classified as a “Husband,” also play an important role, which may reflect differences in household income structure.

Overall, these results suggest that income is influenced by a combination of financial, demographic, and social variables rather than a single dominant factor. This also helps explain why the differences in model performance were relatively small, since multiple features contribute to income prediction rather than one feature driving the outcome on its own.

Discussion

While all three models produced similar accuracy scores, their differences can be better understood by considering how each model approaches the classification task. Radom Forest achieved the highest accuracy, which is expected because it combines multiple decision trees and is able to capture more complex, nonlinear relationships in the data. This allows it to fit the training data more closely while still reducing the overfitting compared to a single decision tree.

In contrast, Logistic Regression is a simpler model that assumes a more linear relationship between the features and the target variable. Despite this limitation, it achieved the highest F1 score, suggesting that it generalizes more evenly across both classes. This indicates that the added complexity of tree-based models does not necessarily improve balanced performance for this dataset.

A major factor influencing all models performance is the class imbalance present in the dataset. Because there are more individuals earning ≤ 50K than >50K, the models are naturally biased toward predicting the majority class. This explains why all models struggle with recall for the higher-income group, even when overall accuracy remains high.

Another challenge is that predicting income is inherently difficult because it depends on a combination of factors that may interact in complicated ways. While models like Random Forest are designed to capture these interactions, the limited improvement in performance suggests that the separation between income classes is not especially distinct based on the available features.

Ultimately, the results show that increasing model complexity leads to only marginal improvements. This suggests that the structure of the data, rather than the choice of model, is the main factor limining performance in this classification task.

Conclusion

In summary, this project showed that all three machine learning models were able to predict income level with fairly similar performance. Random Forest achieved the highest accuracy, while Logistic Regression had the strongest F1 score, making it the most balanced model for this task. Even though the more complex models were able to capture additional patterns in the data, the improvement over Logistic Regression was still pretty small.

The results also showed that predicting individuals in the higher-income class was more difficult across all three models. This was mainly due to class imbalance in the dataset, along with the fact that income is influenced by several overlapping factors rather than one simple pattern. The feature importance results supported this as well, showing that variables such as marital status, education level, capital gain, age, and hours worked per week all played a role in prediction.

Overall, this analysis shows that machine learning can be effective for identifying general income trends, but it also highlights the limits of prediction when working with complex real-world data.